

Summary

AI Engineer and Software Developer building intelligent systems that go beyond demos into real-world use. Combines RAG pipelines, computer vision, and GenAI with production-grade full-stack engineering to ship applications that are reliable and user-focused.

Experience

AI Engineer Intern

Jan 2026 – Present

Global Infoventures Pvt. Ltd.

Delhi, India

- Engineered an automated attendance marking system using computer vision-based face detection, achieving **98% recognition accuracy** in pilot deployment.
- Built internal **RAG pipelines** supporting bilingual (Hindi/English) document querying and QA, improving retrieval relevance by **~40%** through optimized chunking and reranking strategies.
- Authored Jupyter-based technical curriculum on model fine-tuning, computer vision, and GPU-accelerated computing, adopted into the company's **TCS joint-venture LMS** platform serving **2,00,000+ students**; developed and benchmarked AI services on **NVIDIA A100/H200** infrastructure for production deployment.

Projects

Real-Time Multi-Object Tracking Pipeline [\[GitHub\]](#) | RT-DETR, ByteTrack, OpenCV, Pandas, PyYAML

- Built an end-to-end tracking pipeline using **RT-DETR-R50** and **ByteTrack** for Indian road environments, achieving stable persistent IDs across video streams with **92% tracking accuracy (MOTA)**.
- Processed **50+ hours of video** in batch mode, producing annotated outputs, side-by-side comparisons, and per-detection **CSV logs**, with pipeline behavior fully configurable via YAML.

Leaf Disease Detection System [\[GitHub\]](#) | Python, U2-Net, ResNet, Flask, React, OpenCV, NLP

- Built a hybrid deep learning model (**U2-Net + ResNet**) trained on the **PlantVillage dataset (~60,000 images)**, achieving **96% classification accuracy** and reducing misclassification rate by **~15%**.
- Extended into a full-stack web app (**React + Flask**) with image upload, bilingual results, severity scoring, and an **LLM-powered chatbot** for real-time farmer queries.

Clarity – Modern Guidance, Grounded in Wisdom [\[GitHub\]](#) [\[Live App\]](#) | FastAPI, React, FAISS, RAG, LLMs, TypeScript

- Built a multilingual **RAG-based guidance platform** that retrieves and reranks Bhagavad Gita verses using **Top-K semantic search** to deliver contextually relevant answers for modern-life queries.
- Developed an end-to-end **multi-agent pipeline** where retrieved verses are processed through reasoning to generate grounded responses with **real-time TTS speech output** across **11 regional languages**.
- Improved scalability and response reliability by serving static scripture data from the frontend and engineering an async **FastAPI + SSE streaming** backend with background warmup orchestration, cutting average response latency by **~40%**.

Technical Skills

Programming: Python, JavaScript, TypeScript, SQL

AI/ML Core: Machine Learning, Deep Learning, CNNs, Vision Transformers, ResNet, U2-Net, RT-DETR, ByteTrack, Natural Language Processing, Data Preprocessing

GenAI/LLM: RAG Pipelines, LLM Integration, LangGraphs, Prompt Engineering, Model Fine-Tuning, Hugging Face, Vector Databases, FAISS

ML Infra & GPU: PyTorch, NumPy, Pandas, Matplotlib, TensorRT, CUDA, NVIDIA A100/H200

Web & Backend: Flask, FastAPI, React, Node.js, MongoDB, REST APIs

DevOps & Cloud: Git, GitHub, Docker, Jenkins, AWS, Postman, Jira

Design & Other: Figma, Wireframing, Prototyping, Adobe Suite

Education

Bachelor of Technology in Computer Science

Aug. 2022 – May 2026

NIIT University | AI and ML Specialisation

Neemrana, Rajasthan

- Coursework: Machine Learning, Generative AI, AI Agents and Systems, Software Engineering, DBMS

Certifications

Oracle Cloud Infrastructure 2025 Generative AI Professional [\[Certificate\]](#)

Oracle | Generative AI, model deployment on OCI, prompt engineering, enterprise AI integration

Model Context Protocol (MCP) Advanced Development Certificate [\[Certificate\]](#)

Anthropic | MCP architecture, StreamableHTTP/SSE transport, sampling, notifications, file access control, deployment and scaling concepts